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TME–407

B. TECH. (ME) (FOURTH SEMESTER)
MID SEMESTER
EXAMINATION, March, 2024

FLUID MECHANICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a) and (b) in each main question.

(iii) Total marks in each main question are **ten**.

(iv) Each sub-question carries 10 marks.

1. (a) Write short notes on specific weight, specific volume and specific weight.

(CO1)

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OR

(b) A square place 0.1 m side moves parallel to second place with a velocity of 0.1 ms^{-1}, both plates being immersed in water. If the viscous force is 0.002 N and the coefficient of viscosity 0.001 poise, what is the distance between the plates ? (CO1)

2. (a) Write down the rheological classification of fluid. (CO1)

OR

(b) What is the excess pressure inside a spherical drop of radius 1 mm if the surface tension of water is 7.3×10^{-2} N/m ? (CO1)

3. (a) What is Pressure ? Derive an equation for hydrostatic law. (CO2)

OR

(b) The thrust p developed by the propeller is the function of angular velocity w, speed

of advance v, diameter D, dynamic viscosity μ, mass density ρ and the velocity of sound c. Establish a relationship. (CO2)

4. (a) What is Buoyancy ? Write down the condition of stability of floating bodies. (CO2)

OR

(b) The right limb (leg) of a simple U tube manometer containing mercury is open to the atmospheric, while the left limb is connected to a pipe in which a fluid of sp. gr. 0.9 is flowing. The center of the pipe is 12 cm below the level of mercury in the right limb. Find the pressure of fluid in the pipe, if the difference of mercury level in the two limbs is 20 cm. (CO2)

5. (a) What do you understand by geometric, dynamic and kinematic similarity ? (CO2)

OR

(b) What are the different model laws ? (CO2)

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